

Foundation (Jalapeno)

Q1. Factor the trinomial: $x^2 + 5x + 6$

- (A) $(x + 2)(x + 3)$
- (B) $(x + 1)(x + 6)$
- (C) $(x - 2)(x - 3)$
- (D) $(x - 1)(x - 6)$
- (E) $(x + 1)(x - 6)$

Q2. Factor the trinomial: $x^2 - 7x + 12$

- (A) $(x - 3)(x - 4)$
- (B) $(x - 1)(x - 12)$
- (C) $(x + 3)(x + 4)$
- (D) $(x + 1)(x + 12)$
- (E) $(x - 2)(x - 6)$

Q3. A stock's price is modeled by $p(t) = t^2 + 4t + 4$. Factor $p(t)$

- (A) $(t + 2)(t + 2)$
- (B) $(t - 2)(t - 2)$
- (C) $(t + 1)(t + 4)$
- (D) $(t - 1)(t - 4)$
- (E) $(t + 3)(t + 1)$

Q4. A company's profit is modeled by $P(x) = x^2 + 9x + 20$. Factor $P(x)$

- (A) $(x + 4)(x + 5)$
- (B) $(x - 4)(x - 5)$
- (C) $(x + 1)(x + 20)$
- (D) $(x - 1)(x - 20)$
- (E) $(x + 2)(x + 10)$

Q5. Factor the trinomial: $x^2 - 2x - 15$

- (A) $(x - 5)(x + 3)$
- (B) $(x + 5)(x - 3)$
- (C) $(x - 1)(x - 15)$
- (D) $(x + 1)(x + 15)$
- (E) $(x - 2)(x - 7)$

Q6. Factor the trinomial: $x^2 + x - 12$

- (A) $(x + 4)(x - 3)$
- (B) $(x - 4)(x + 3)$
- (C) $(x + 1)(x - 12)$
- (D) $(x - 1)(x + 12)$
- (E) $(x + 2)(x - 6)$

Q7. A tax model is given by $T(x) = x^2 + 2x - 15$. Factor $T(x)$

- (A) $(x + 5)(x - 3)$
- (B) $(x - 5)(x + 3)$
- (C) $(x + 1)(x - 15)$
- (D) $(x - 1)(x + 15)$
- (E) $(x + 2)(x - 7)$

Q8. Factor the trinomial: $x^2 - 9x + 20$

- (A) $(x - 4)(x - 5)$
- (B) $(x + 4)(x + 5)$
- (C) $(x - 1)(x - 20)$
- (D) $(x + 1)(x + 20)$
- (E) $(x - 2)(x - 10)$

Open Ended Questions (FRQ)

Q9. Factor the trinomial $x^2 + 10x + 24$. Show all steps.

Q10. A savings account's interest is modeled by $I(x) = x^2 + 6x + 8$. Factor $I(x)$ and explain the result in the context of savings.

Chef's Tips

- Emphasize factoring as a crucial step in solving quadratic equations
- Encourage students to use real-world applications to understand the relevance of factoring
- Chef's Tips